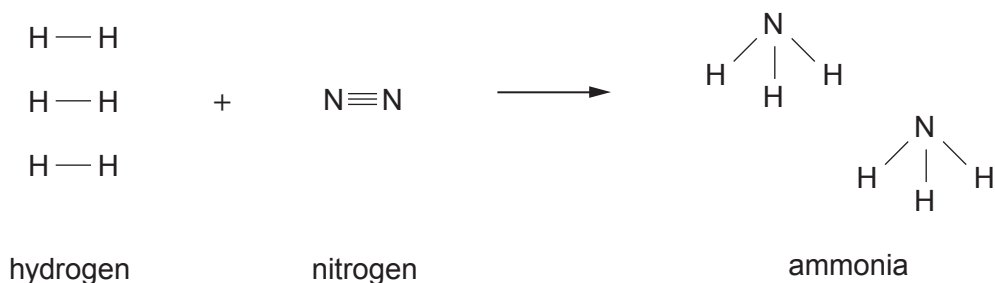


8. Ammonia is manufactured from hydrogen and nitrogen in the Haber process.

- (a) The equation shows the bonds which are broken and the bonds which are formed in the production of ammonia.

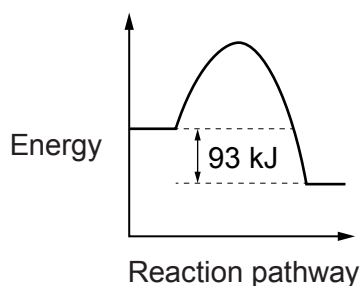


The bond energy of a H—H bond is 436 kJ. The energy needed to break **all** the bonds in the **reactants** is 2253 kJ.

- (i) Calculate the energy needed to break a N≡N bond. [2]

Energy = kJ

- (ii) The diagram below is the energy profile for the reaction.



Calculate the energy released when **all** the bonds in the ammonia molecules are formed. [1]

Energy = kJ



Examiner
only

- (b) The table below shows the percentage yield of ammonia under different pressure and temperature conditions.

Pressure (atm)	Temperature (°C)			
	200	300	400	500
	Percentage yield of ammonia			
100	81.7	52.5	25.2	10.6
200	89.0	66.7		18.3
400	94.6	79.7	55.4	31.9
1000	98.3	92.6	79.8	57.5

- (i) Describe the relationship between the percentage yield of ammonia and the temperature.

[1]

- (ii) **Circle** the most likely percentage yield of ammonia if the process were carried out at 200 atm and 400 °C.

[1]

20 % 30 % 40 % 50 %



- (c) A student carries out a series of chemical tests on solutions of three unknown compounds, **A**, **B** and **C**. Her results are recorded in the table.

Compound	Flame test	Add NaOH(aq)	Add HCl(aq)	Add BaCl ₂ (aq)	Add AgNO ₃ (aq)
A	no colour	pungent gas given off turns damp red litmus paper blue	no reaction	no reaction	white precipitate forms
B	green flame	blue precipitate forms	no reaction	white precipitate forms	no reaction
C	lilac flame	no reaction	gas given off turns limewater milky	no reaction	no reaction

Use the information provided to identify compounds **A**, **B** and **C**.

[3]

A

B

C

END OF PAPER



FORMULAE FOR SOME COMMON IONS

POSITIVE IONS		NEGATIVE IONS	
Name	Formula	Name	Formula
aluminium	Al^{3+}	bromide	Br^-
ammonium	NH_4^+	carbonate	CO_3^{2-}
barium	Ba^{2+}	chloride	Cl^-
calcium	Ca^{2+}	fluoride	F^-
copper(II)	Cu^{2+}	hydroxide	OH^-
hydrogen	H^+	iodide	I^-
iron(II)	Fe^{2+}	nitrate	NO_3^-
iron(III)	Fe^{3+}	oxide	O^{2-}
lithium	Li^+	sulfate	SO_4^{2-}
magnesium	Mg^{2+}		
nickel	Ni^{2+}		
potassium	K^+		
silver	Ag^+		
sodium	Na^+		
zinc	Zn^{2+}		



2
1

Group



4

50

6

1

0

1 H Hydrogen 1																																																																						
7	Li	3	9	Be	4	Beryllium																																																																
23	Na	11	24	Mg	12	Magnesium																																																																
39	K	19	40	Ca	20	Calcium	45	Sc	21	Scandium	48	Ti	22	Titanium	51	V	23	Vanadium	52	Cr	24	Chromium	55	Mn	25	Manganese	56	Fe	26	Iron	59	Co	27	Cobalt	59	Ni	28	Nickel	63.5	Cu	29	Copper	65	Zn	30	Zinc	70	Ga	31	Gallium	73	Ge	32	Germanium	75	As	33	Arsenic	79	Se	34	Selenium	80	Br	35	Bromine	84	Kr	36	Krypton
86	Rb	37	88	Sr	38	Strontium	89	Y	39	Yttrium	91	Zr	40	Zirconium	93	Nb	41	Niobium	96	Mo	42	Molybdenum	99	Tc		Technetium	101	Ru	44	Ruthenium	103	Rh	45	Rhodium	106	Pd	46	Palladium	108	Ag	47	Silver	112	Cd	48	Cadmium	115	In	49	Indium	119	Sn	50	Tin	122	Sb	51	Antimony	128	Te	52	Tellurium	127	I	53	Iodine	131	Xe	54	Xenon
133	Cs	55	137	Ba	56	Barium	139	La	57	Lanthanum	179	Hf	72	Hafnium	181	Ta	73	Tantalum	184	W	74	Tungsten	186	Re	75	Rhenium	190	Os	76	Osmium	192	Ir	77	Iridium	195	Pt	78	Platinum	197	Au	79	Gold	201	Hg	80	Mercury	204	Tl	81	Thallium	207	Pb	82	Lead	209	Bi	83	Bismuth	210	Po	84	Polonium	210	At	85	Astatine	222	Rn	86	Radon
223	Fr	87	226	Ra	88	Radium	227	Ac	89	Actinium																																																												

Key

relative atomic mass

$$A_r$$

Symbol

Name

atomic number